2026 SHENZHEN FINTECHATHON · INTERNATIONAL TRACK · TOPIC C

TuitionPilot

An AI agent that helps an Indian family pay university tuition abroad, and pays only after the parent’s device signs the exact plan.

Today a family works out four things alone. One mistake costs money.

- How much of the yearly **LRS limit** is left?
- How much **TCS** tax will the bank add?
- Which route is **really cheapest** once the hidden markup counts?
- Which **papers** does the bank want?

Chandan and Nishanth · Synthetic data only · No real money · Not tax advice

Product: a guided six-step story

1
Welcome
Two consents before any data is read

2
Invoice
Animated check of the signature chain

3
Questions
One at a time; asks instead of guessing

4
Best route
Cost-on-top chart, what-if sliders, LRS gauge

5
Review & sign
PIN pad; the parent’s device signs the plan

6
Receipt
Transaction, certificate, signature, audit entry

Agent, live panel: every step and policy decision as it happens, the rule trace with sources, what the AI saw (masked), the audit chain.

Attack lab: runs 16 attacks against a separate copy of the agent and shows which defence stopped each. **Parent and student sign-in:** a student can review but never sign.

Behind the screens: Intake → Mask → Extract → Validate → Rules → Compare → Plan → Policy gate → Approve → Execute (sandbox) → Record → Track

Code computes; the model explains. The model sees masked text only, and any number it writes must match the engine. Today it is an offline stand-in.

FOR BANKERS AND ACADEMICS

LRS and TCS: the rules the engine uses

Rule	Value in the rule pack	From	Status
LRS yearly cap	USD 250,000 per person, April to March, all purposes	not set	Unverified
TCS threshold	Rs 10 lakh per PAN per year, all LRS purposes	1 Apr 2025	Secondary source
Education, self-funded	2% on the part above the threshold (was 5%)	1 Apr 2026	Secondary source
Education, loan-funded	0% (loan from a notified institution)	1 Apr 2025	Secondary source
No PAN	5% on the part above the threshold	not set	Unverified
Earlier other-purpose payments	Use up the Rs 10 lakh threshold	not set	Unverified
TCS base	Amount sent, not fees	not set	Unverified

TCS = rate × (this year's total − the larger of earlier payments or Rs 10 lakh), never below zero. Loan-funded: zero.

Rounded to the whole rupee. The engine refuses a date before a rule took effect instead of guessing.

FOR BANKERS AND ACADEMICS

Worked examples, step by step

A2 · Self-funded Rs 15 lakh, no earlier payments

This year’s total	Rs 15,00,000
Minus the threshold	– Rs 10,00,000
Taxable part	Rs 5,00,000
TCS at 2%	**Rs 10,000**

Same Rs 15 lakh paid with an education loan (A3): TCS Rs 0.

A8 · USD 20,000 via Provider B at 90.20

Amount sent	Rs 18,04,000
TCS at 2% on Rs 8,04,000	+ Rs 16,080
Provider fee	+ Rs 3,000
Total leaving the account	**Rs 18,23,080**

Rs 8,04,000 is the part of Rs 18,04,000 above Rs 10 lakh.

A1 Rs 8 lakh: TCS nil

A4 Rs 6 lakh earlier + Rs 7 lakh: Rs 6,000

A5 Rs 12 lakh other + Rs 5 lakh: Rs 10,000 (unverified rule)

A6 No PAN: asks for it first; 5% = Rs 25,000 (unverified rule)

Route comparison: why Provider B wins

USD 20,000	Provider A	Provider B
Rate (INR per USD)	90.50	90.20
Amount sent	18,10,000	18,04,000
Provider fee	1,500	3,000
TCS at 2%	16,200	16,080
Total (Rs)	18,27,700	**18,23,080 cheapest**

Provider A has the lower fee, but B is Rs 4,620 cheaper overall.

In the app, the Best route screen shows this as a cost-on-top chart, with what-if sliders and an LRS limit gauge.

The “why”, line by line

Exchange rate: B saves	+ Rs 6,000
Provider fee: B costs more	– Rs 1,500
TCS: B saves	+ Rs 120
Saving with B	**Rs 4,620**

Code writes every line, and the lines add up exactly to the saving. Each line lists the rules it used and which are unverified.

HIDDEN MARKUP

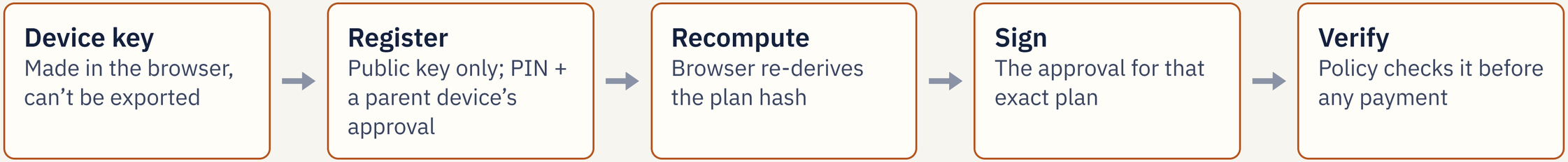
Against a mid-market rate of 90.00, Provider A’s 0.50 gap × USD 20,000 = **Rs 10,000 (0.56%)** built into the rate, never shown as a fee. Provider B’s gap: Rs 4,000.

Security: signatures instead of lists

University invoice. A changed bank account breaks the signature.



Parent approval. The agent never holds the parent's key, so it can't forge an approval or change the plan.



T0 Read, check signatures · none	**T1** Compute, draft · none	**T2** Save, share drafts · click	**T3** Sandbox payment · PIN + device signature	**T4** New payee, change limits, skip approval · never

Ed25519 in plain Python following RFC 8032, nothing to install, checked against the RFC test vectors and a browser-made signature. The model has no payment tool.

Built for families: sign-in, vault, next action

Passwordless sign-in

Each device signs a one-time challenge with a key that never leaves the browser. Roles are fixed at sign-in: a student can never act as the parent.

New devices need two locks

The parent’s PIN **and** a signature from an enrolled parent device over the exact request. Knowing the PIN alone no longer adds a device.

Encrypted family vault

Admission letters and receipts are locked in the browser: AES-256-GCM, with a key wrapped for each allowed device via X25519. The server holds ciphertext only.

Asks once, then resumes

A missing due date gets one targeted question. The agent picks its next action from the current state and never re-runs a finished check. Reads PDF, JSON or text invoices.

Source: app/family.py · agent/agent.py (next_action) · agent/pdf.py · docs/TECHNICAL.md 4.7 · 7 / 9

Evidence: 27 scenarios, one command

27 / 27
scenarios pass

227
tests pass

16 / 16
attacks stopped in the lab

153
audit entries, chain verifies

Scenarios by group

A Task completion	10 / 10 pass
B Safe execution	6 / 6 pass
C Attacks, incl. forged approval, revoked key	9 / 9 pass
D Privacy	2 / 2 pass

Finance audit: 400 random cases

The engine matched an independent formula in every case. The audit found 3 agent bugs, now fixed with tests:

- An earlier in-app payment was not counted
- Inconsistent USD answers were accepted
- No PAN + loan applied 0% silently; now a warning

python -m scenarios.runner · python -m app.demo_run → data/evidence/demo/ · run 18 Oct 2026 · 8 / 9

Limits, stated plainly

Tax rules not yet verified

- Unverified: LRS cap USD 250,000; no-PAN rate 5%; earlier other-purpose payments counting toward Rs 10 lakh; TCS on the amount sent, not fees; GST rate per provider
- Secondary sources only: the Rs 10 lakh threshold, the 2% and 0% education rates
- Not modelled: GST on the currency conversion itself

Every output that uses one of these shows an “unverified” tag. To check against the RBI Master Direction and Income-tax Act 2025 s.394 before final submission.

Other known limits

- The AI model is an offline, deterministic stand-in; no hosted model yet
- Our Ed25519 code is not hardened: not constant-time, not audited
- Demo keys are synthetic; the trust root is a single point of trust
- Addresses and unlabelled names are not masked
- Scanned PDF invoices can’t be read: no OCR
- Demo PIN is fixed for judges
- Audit log shows tampering but can’t stop a full rewrite

Synthetic data only · No real money · Drafts, not official RBI or bank forms · Not tax advice

Source: docs/SECURITY_SELF_ASSESSMENT.md, known risks R1–R26 · docs/TECHNICAL.md 7 · 9 / 9